

1CHEM 1020-001

**INTRODUC
TORY
CHEMISTR
Y II
Fall
2024**

Instructor: Dr. Tammy Melton

3135

Office: SCI

Lecture: MWF 9:10-10:05

Location: WPS 310

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Office Hours: MWR 2-5; T 2-4; F 2-4; (and by appointment)

Chem 1020 - This course is a continuation of CHEM 1010 - Introductory General Chemistry I. Topics will include: hydrocarbons, organic functional groups, isomerism, carbohydrates, lipids, nucleic acids, proteins, and enzymes. Applications to aspects of human health will be highlighted when appropriate.

Pre-requisite: CHEM 1010 and lab, or equivalent.

Co-requisite: Students **MUST** be registered in a section of Chem 1020 Laboratory concurrent with the lecture. Students must **PASS** the laboratory component in order to pass the course. One grade will be assigned for the combination of lab and lecture.

Required Text: Our text is *Chemistry for Today: General, Organic, and Biochemistry* by Seager and Slabaugh, 9th Edition, Thomson/Brooks-Cole 2018, ISBN 978-1-305-96006-0. There is a lot of practice homework in the book – you need it.

Laboratory Text: *Introductory Chemistry 2 Laboratory Manual, Chemistry 1021* by Tammy J. Melton, Hayden-McNeil Publishing, 2023. ISBN 978-1-5339-5616-3.

Equipment: Goggles for lab are **absolutely required**. You will also need a calculator for one or two lab experiments, but a calculator will not be needed for lecture. A molecular model kit may be helpful and is recommended, but is not required.

Tentative Grading Procedure:

Laboratory Grades: We will perform 11 experiments, which count 10 points each. I will drop the lowest grade so that equals 100 points. *If you miss an experiment, that is the grade you drop; there are no makeup labs.* There will be 2 lab exams counting 50 points each. The total possible points in lab is therefore 200.

Homework is the only way to master the material. A copy of the recommended homework problems from the textbook is attached as the last page. You must work homework problems diligently. You *cannot* wait until the day before an exam and then try to work them all. If you have not mastered previous material, you will not be able to keep up with new material.

Lecture Grades: There will be five regular exams at 100 points each (total 500 points). EACH EXAM IS BY NATURE, AND WILL BE BY DESIGN, **COMPREHENSIVE**. The final exam is comprehensive and will be worth 100 points.

Quizzes There are six online quizzes, distributed throughout the semester, intended to help you prepare for exams. They are worth 20 points each. You can drop the lowest grade, so best five of six quizzes counts a total of 100 points, *the same as a lecture exam*. You must complete the quizzes no later than 9:00 am on the days indicated on the syllabus (generally exam days). You are free to use whatever tools you want to answer the questions: your notes, textbook, model kits, instructor, fellow students, the internet. Consider this as a check on your homework.

Therefore, the overall total of points is:

Laboratory
200 points
5 Exams @ 100 points each
500 points
Best five of six quizzes
100 points
Comprehensive Final Exam
100 points

Total 900 points

The grade earned will be assigned based on *points* not percentages; the *approximate* percentages corresponding to the points is indicated for your information. These cutoffs will be followed.

Grade	A	B	C	D	F
pts:	900 - 810	809 - 720	719 - 630	629 - 540	539 - 0
	(100 - 90%)	(89 - 80%)	(79 - 70%)	(69 - 60%)	(below 60%)

Attendance: I expect you to attend class every day. It will be very difficult to do well if you do not come. You are responsible for all information covered in class, including changes to this syllabus, whether or not you are present. I give unannounced quizzes worth bonus points at random times. These cannot be made up. There is no other form of “extra credit.”

Missing an Exam In the event that a student must miss an exam for an excusable reason, **it is the student’s responsibility to inform the instructor beforehand**. The only reasonable exception to this is if you are delayed by car failure or accident on the way to class. In such case the student should TELEPHONE the instructor 615-898-2626 as soon as possible to explain. Late excuses two or three days (or weeks!) after an exam has occurred will be met with little sympathy. **Failure to discuss with the instructor any missed exam in a timely manner will likely result in the exam NOT being excused.**

Deportment

1. Please make every effort to arrive and be ready to start class **on time**.
2. Since our class meets early, and we all have busy schedules, feel free to bring a soda, coffee, or other *discreet* snack to class. Please clean up any mess you make.
3. We will respect each other. Please don't talk while I am talking. In return, I promise to give your questions every courtesy. If my answer is insufficient, please *ask again*.
4. My office hours are for you, please come by with any questions or problems. I am eager to help you in any way possible. If you are not free during my scheduled office hours, we can arrange a mutually convenient time—just ask. This is my job; this is what you pay for.
5. *Cell phone use, text messaging, I-Pad, laptop use, and other electronic communication while class is underway is rude and will not be tolerated.* Use of electronic communication devices during an exam is considered a form of cheating, and will result in failure of the course.

Cheating: No cheating of any kind will be tolerated. Any student caught cheating in this class or in the lab will receive **a grade of F in the course**. Academic dishonesty includes but is not limited to: copying from another student during a quiz, exam, or lab report; using crib notes during a quiz or exam; copying or using old laboratory reports for a laboratory experiment; presenting false excuses to obtain privilege for makeup work; and plagiarism.

Students with disabilities: Any student in this class who has a disability must self-identify to the instructor. We will work together to make reasonable accommodations to permit the student to complete the work. You may also contact Disabled Student Services at 898-2783.

THIS SYLLABUS IS NOT A CONTRACT.

Tentative Schedule of Course Topics – The instructor retains the right to make schedule changes as are deemed necessary.

<u>Date</u>	<u>Topic</u>
8-26	Date Topic Introduction, Chapter 11 10-21
8-28	EXAM III(C 15&16) Chapter 11 (Hydrocarbons) 10-23 Chapter 17 (Carbs) Chapter 11
8-30	10-25 Chapter 17 Labor Day - No Class 10-28
9-2	

9-4	Chapter 17 Chapter 11
9-6	10-30 Chapter 18 (Lipids) Chapter 12 (Alkenes, Alkynes) 11-1
9-9	Chapter 18
	Chapte r 12
	11-4
	Chapte r 18
9-11	Chapter 12
9-13	11-6 Chapter 19 (Proteins) Chapter 12
	11-8 Chapter 19
9-16	EXAM I (Chps 11&12) 11-11 Chapter 19

9-18			Chapter 13 (Alcohols)
			11-13 EXAM IV (17,18,19)
9-20			Chapter 13
			11-15 Chapter 20 (Enzymes)
9-23			Chapter 13
			11-18 Chapter 20 Chapter 13
9-25			
			11-20 Chapter 20 Chapter 14 (Aldehydes, Ketones)
9-27			11-22
			Chapter 21 (DNA)
9-30			Chapter 14
			11-25 Chapter 21 EXAM II (Ch 13&14)
10-2			11-27
			Thanksgiving - no class
10-4	Chapter 15 (Carboxylates)	11-29	Thanksgiving - no class
10-7			Chapter 15
			12-2

10-9		Chapter 21 Chapter 15
		12-4 Exam V (Ch 20, 21)
10-11	Chapter 16 (Amines, Amides)	
10-14		Fall Break - No Class 12-11
10-16		FINAL EXAM Chapter 16
10-18		Wednesday 10:00-12:00 Chapter 16

Comprehensive

Q1-Q6 Online (D2L) must be completed no later than 9:00 am on the day of the related exam.

n.b. Sept 8 - last day to drop without a grade
Oct 30 - last day to drop with a "W"

Suggested Homework Problems for Chem 1020 Seager and Slabaugh 9th edition

Chapter 11:

2, 4, 6, 8, 12, 17, 18, 20, 22, 24, 26, 27, 28, 30, 32, 33, 34, 36, 38, 40, 42, 44, 46, 48, 49,
54, 56, 58, 60, 62, 69, 73, 80, 81, 82, 84

Chapter 12:

1, 2, 3, 4, 6, 9, 11, 15, 16, 18, 20, 22, 23, 24, 25, 26, 28, 30, 40, 42, 43, 44, 47, 48, 50, 52, 54, 56,
60, 62, 81, 82, 83

Chapter 13:

3, 4, 5, 6, 7, 10, 12, 14, 15, 16, 17, 18, 19, 20, 22, 23, 24, 26, 28, 30, 36, 40, 42, 43, 44, 45, 47,
48, 56, 64, 72, 74, 75, 76

Chapter 14:

1, 4, 6, 8, 9(there are seven), 11, 12, 15, 16, 18, 20, 22, 24, 28, 31(a-d), 32, 34, 42(a,d,e), 43(a,b,d), 44, 49, 58, 63, 67, 68

Chapter 15:

1, 2, 3, 4, 6, 7, 8, 10, 13, 16, 18, 19, 23, 24, 26, 28, 32, 33, 34, 36(b,c), 38, 40, 47, 48, 50, 51, 54, 58, 63, 66, 73, 75

Chapter 16:

1, 2, 4, 5, 8, 10, 11, 12, 13, 14, 16, 18, 20, 21, 24, 26(a,b,d,e), 28, 32, 34, 38, 42, 44, 46, 48, 50, 52, 56, 66, 72, 73

Chapter 17:

1, 2, 3, 4, 5, 6, 8, 12, 15, 18, 20, 22, 24, 25, 28, 31, 32, 36, 38, 42, 43, 45, 51, 54, 55, 61, 64, 66, 70, 71, 72, 73, 75, (also go back to Chapter 14 and do 14.36, 14.37, 14.38)

Chapter 18:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17, 18, 19, 20, 23, 24, 26, 28, 34, 43, 44, 46, 49, 52, 53, 55, 56, 60, 64, 69, 70, 74, 75, 78, 79, 80, 82

Chapter 19:

2, 4, 6, 9, 11, 16, 18, 20, 22, 23, 27, 28, 30, 32, 34, 35, 36, 37, 38, 40, 41, 43, 44, 45, 46, 47, 48, 49, 50, 52, 53, 63, 65, 66, 70, 71, 72, 73, 74, 76, 77

Chapter 20:

1, 2, 3, 4, 9, 10, 11, 12, 14, 15, 16, 18, 19, 20, 21, 22, 27, 28, 29, 33, 34, 39, 40, 43, 45, 48, 55, 59, 60, 63, 66, 68, 72, 73, 74, 75

Chapter 21:

1, 2, 3, 4, 5, 6, 8, 10, 11, 12, 14, 16, 17, 20, 26, 28, 30, 31, 32, 34, 37, 40, 43, 48, 50, 60, 61, 69, 70, 72, 73, 74, 75, 76